National University of Computer and Emerging Sciences
Islamabad Operating System Lab Spring 2025

Lab Task Threads

LAB TASK 01: Email Parser

Create four threads besides the main thread:

1. **Input Thread**: Takes an email address as input.
2. **Username Extractor Thread**: Extracts and prints the username (before @).
3. **Domain Extractor Thread**: Extracts and prints the domain (after @).
4. **Validation Thread**: Checks if email format is valid (must contain exactly one '@' and a . after it).

All threads execute after input is ready.
Input thread should handle:

- Empty input
- Missing @ or .
- More than one @

Lab Task 2: Multi-Threaded Library Book Issuing System

Scenario:

You're simulating parallel book issuing in a library system. The goal is to handle book issue requests concurrently using multiple threads and calculate the total fine for each student.

Requirements:

1. **Define a `struct BookIssue`** with the following attributes:
    - `issueID`
    - `studentName`
    - `bookTitle`
    - `daysLate`
    - `finePerDay`
2. **Ask the user** to enter the number of book issue records (between **8 to 10**).
3. **Divide the records evenly among 4 threads** (each thread will handle 2 or 3 records).
4. For each record, each thread should:
    - Calculate the **total fine**:
      `totalFine = daysLate * finePerDay`
    - Flag issues with **daysLate > 7** as "**Severely Late**".
    - Print a **summary** of the issue:
        - `Student Name`
        - `Book Title`
        - `Total Fine`
        - Whether it is **Severely Late**
5. After all threads complete, the **main thread should display the total fines collected** by the library.

LAB TASK 03: Multi-Threaded Grocery Bill Calculator

Create five threads including the main thread:

1. **Input Thread**: Takes the number of grocery items and their prices as input.
2. **Total Bill Thread**: Calculates the total bill.
3. **Discount Thread**: Applies a 10% discount if total bill is above 100.
4. **Tax Thread**: Adds 5% sales tax to the total after discount (if applicable).
5. **Item Count Thread**: Counts how many items were bought.

- The **Main Thread** displays the final bill with tax and discount applied.

LAB TASK 04: String Reverser and Inspector

Create four threads besides the main thread:

1. **Input Thread**: Takes a string from the user.
2. **Reverser Thread**: Reverses the string and prints it.
3. **Palindrome Checker Thread**: Checks if the string is a palindrome.
4. **Character Type Counter Thread**: Counts:
    - Alphabets

- Digits
- Special characters

No processing thread should run before input is given.
Input thread must reject strings with only spaces.

LAB TASK 05: Student Record Analyzer

Create four threads besides the main thread:

1. **Input Thread**: Takes a student's full record as input:
   - Name (string), Roll Number (string), CGPA (float)
2. **Name Formatter Thread**: Converts name to uppercase.
3. **CGPA Evaluator Thread**: Evaluates performance:
   - "Excellent" if CGPA ≥ 3.5
   - "Good" if CGPA ≥ 3.0
   - "Average" if CGPA ≥ 2.5
   - "Poor" otherwise
4. **Roll Number Validator Thread**: Checks if the roll number format is valid (e.g., must start with digits, then letters like "22K-1234")

All threads run only after the input thread completes.

THE END
